IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

Distributed Optimisation and Learning

Thursday 4 May 2023 10:00 am

Time allowed: 3:00 hours

There are a total of 4 problems that total to 100 points.

This exam is OPEN BOOK.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Vlaski
Second Marker(s) : K. Leung

© Imperial College London

Question 1. The following are true/false questions. Full credit will only be given for correct answers with correct justification, but partial credit is possible.

(a) [5 points] (true/false) A collection of K agents always exhibits a K-fold improvement in steady-state performance when collaborating through the Consensus+Innovations algorithm.

(b) [5 points] (true/false) Consider a network of K agents described by an adjacency matrix $A \in \mathbb{R}^{K \times K}$. Suppose the matrix is symmetric and doubly-stochastic, i.e., $A\mathbb{1} = \mathbb{1}$ and $A = A^{\mathsf{T}}$. It then follows that $\lim_{i \to \infty} A^i$ exists and is a rank-one matrix.

(c) [5 points] (true/false) Suppose a network of highly heterogeneous agents is sparsely connected. For moderate step-sizes and gradient approximation accuracies, it can then be expected that primal-dual algorithms for decentralized optimization will outperform penalty-based algorithms.

(d) [5 points] (true/false) All else being equal, the Consensus+Innovations algorithm implemented over a more densely connected network will perform better than over a sparse network.

Question 2. Each agent in a network has access to a local objective $J_k(w)$ with $w \in \mathbb{R}^M$. At each iteration, each agent picks a random coordinate $\boldsymbol{m}_{k,i}$ by sampling uniformly from the set of integers between 1 and M. It then constructs local gradient approximations by differentiating only relative to the $\boldsymbol{m}_{k,i}$-th entry:

$$\widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1}) \triangleq M \frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[\boldsymbol{m}_{k,i}]} e_{\boldsymbol{m}_{k,i}} \tag{1}$$

(a) [15 points] Show that this gradient approximation satisfies the gradient noise conditions:

$$\mathbb{E}\left\{\widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1})|\boldsymbol{w}_{k,i-1}\right\} = \nabla J_k(\boldsymbol{w}_{k,i-1}) \tag{2}$$

$$\mathbb{E}\left\{\|\nabla J_k(\boldsymbol{w}_{k,i-1}) - \widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1})\|^2|\boldsymbol{w}_{k,i-1}\right\} \leq \beta_k^2\|w_k^o - \boldsymbol{w}_{k,i-1}\|^2 + \sigma_k^2 \tag{3}$$

for suitably chosen constants β_k^2, σ_k^2.

(b) [10 points] Write down the Exact Diffusion recursions for this particular choice of gradient approximation. How do you expect the limiting error $\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ to compare to the implementation where you use true gradients $\nabla J_k(w)$ at each agent?

Question 3. We consider a network of K agents, each with local objective:

$$J_k(w) = \frac{1}{2}\mathbb{E}\Big(\boldsymbol{\gamma}_k - \boldsymbol{h}_k^{\mathsf{T}} w\Big)^2 \tag{4}$$

The agents employ the diffusion algorithm with locally constructed mini-batch approximations of the gradient:

$$\boldsymbol{\psi}_{k,i} = \boldsymbol{w}_{k,i-1} + \frac{\mu}{B}\sum_{b=1}^{B} \boldsymbol{h}_{k,i,b}\left(\boldsymbol{\gamma}_{k,i,b} - \boldsymbol{h}_{k,i,b}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right) \tag{5}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\psi}_{\ell,i} \tag{6}$$

(a) [10 points] Determine an expression for the limiting performance $\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ in terms of algorithm parameters and data statistics. You can assume the step-size to be small enough, so that any higher-order terms (i.e., $O(\mu^2)$) can be disregarded, and that all agents are homogenous (i.e., their data is i.i.d.).

Moving forward, agents are faced with communication constraints, and hence exchange quantized estimates:

$$\boldsymbol{\psi}_{k,i} = \boldsymbol{w}_{k,i-1} + \frac{\mu}{B}\sum_{b=1}^{B} \boldsymbol{h}_{k,i,b}\left(\boldsymbol{\gamma}_{k,i,b} - \boldsymbol{h}_{k,i,b}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right) \tag{7}$$

$$\boldsymbol{\phi}_{k,i} = \boldsymbol{\mathcal{Q}}_k(\boldsymbol{\psi}_{k,i}) \tag{8}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\phi}_{\ell,i} \tag{9}$$

Where the quantizer satisfies the typical conditions:

$$\mathbb{E}\left\{\boldsymbol{\mathcal{Q}}_k(\boldsymbol{\psi}_{k,i})|\boldsymbol{\psi}_{k,i}\right\} = \boldsymbol{\psi}_{k,i} \tag{10}$$

$$\mathbb{E}\left\{\left\|\boldsymbol{\psi}_{k,i} - \boldsymbol{\mathcal{Q}}_k(\boldsymbol{\psi}_{k,i})\right\|^2|\boldsymbol{\psi}_{k,i}\right\} \le \beta_q^2\|\boldsymbol{\psi}_{k,i}\|^2 + \sigma_q^2 \tag{11}$$

with common parameters β_q^2, σ_q^2.

(b) [10 points] Determine an expression for the limiting performance $\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ in terms of algorithm parameters, data statistics and quantizer parameters.

(c) [10 points] Suppose each agent employs the same quantization scheme with T bits, and the quantization parameters scale as:

$$\beta_q^2 = c_1 2^{-T} \tag{12}$$

$$\sigma_q^2 = c_2 2^{-T} \tag{13}$$

Determine a sufficient condition on μ and T to ensure:

$$\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2 \le \epsilon \tag{14}$$

Question 4. We will examine a random variant of the static consensus averaging algorithm, where at any point in time, agents employ a random weight matrix:

$$\boldsymbol{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} \boldsymbol{a}_{\ell k,i} \boldsymbol{w}_{\ell,i-1} \tag{15}$$

Each agent is initialized at some deterministic local signal g_k, i.e., $\boldsymbol{w}_{k,0} = g_k$. The random combination weights are collected in a random matix $\boldsymbol{A}_i$ that satisfies almost surely:

$$\boldsymbol{A}_i = \boldsymbol{A}_i^{\mathsf{T}} \tag{16}$$

$$\boldsymbol{A}_i \mathbb{1} = \mathbb{1} \tag{17}$$

Furthermore, the weight matrices are identically and independently distributed over time with mean and variance:

$$\mathbb{E}\boldsymbol{A}_i = \overline{A} \tag{18}$$

$$\mathbb{E}\left\|\boldsymbol{A}_i - \overline{A}\right\|^2 = \sigma_A^2 \tag{19}$$

and we assume $\overline{A}$ to be a primitive matrix with mixing rate $\lambda_2 = \rho\left(\overline{A} - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\right) < 1$.

(a) [10 points] Determine an expression for the evolution of the network centroid:

$$\boldsymbol{w}_{c,i} \triangleq \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{w}_{k,i} \tag{20}$$

(b) [15 points] Determine an expression for the evolution of mean estimates:

$$\overline{w}_{k,i} \triangleq \mathbb{E}\boldsymbol{w}_{k,i} \tag{21}$$

Do they converge? If so, what is their limiting point and rate of convergence?